



Linux classes for
device and drivers

Linux Drivers Character Device Files Creation and Operations

Part—5

This article is a continuation of the series on Linux device drivers, and carries on the discussion on character drivers and their implementation.

In my previous article, I had mentioned that even with the registration for the `<major, minor>` device range, the device files were not created under `/dev`—instead, Shweta had to create them manually, using `mknod`. However, on further study, Shweta figured out a way to automatically create the device files, using the `udev` daemon. She also learnt the second step to connect the device file with the device driver—linking the device file operations to the device driver functions. Here is what she learnt.

Automatic creation of device files

Earlier, in kernel 2.4, the automatic creation of device files was done by the kernel itself, by calling the appropriate APIs of `devfs`. However, as the kernel evolved, kernel developers realised that device files were more related to user-space and hence, as a policy, that is where they ought to be dealt with, not at the kernel. Based on this idea, the kernel now only populates the appropriate device class and

device information into the `/sys` window, for the device under consideration. User-space then needs to interpret it and take appropriate action. In most Linux desktop systems, the `udev` daemon picks up that information, and accordingly creates the device files.

`udev` can be further configured via its configuration files to tune the device file names, their permissions, their types, etc. So, as far as the driver is concerned, the appropriate `/sys` entries need to be populated using the Linux device model APIs declared in `<linux/device.h>`. The rest should be handled by `udev`. The device class is created as follows:

```
struct class *cl = class_create(THIS_MODULE, "device class
name");
```

Then, the device info (`<major, minor>`) under this class is populated by:

```
device_create(cl, NULL, first, NULL, "device name format", ...);
```